

Multiple Choice (10 marks)

1. Which nuclide has an NMR frequency of 115.5 MHz in a 20.0 T magnetic field?
 - (a) ^{17}O
 - (b) ^{19}F
 - (c) ^{29}Si
 - (d) ^{31}P
2. Cobalt-60 is used in the radiation therapy of cancer and can be produced by bombardment of cobalt-59 with which of the following?
 - (a) Neutrons
 - (b) Alpha particles
 - (c) Beta particles
 - (d) X-rays
3. The normal modes of a carbon dioxide molecule that are infrared-active include which of the following? I. Bending II. Symmetric stretching III. Asymmetric stretching
 - (a) I only
 - (b) II only
 - (c) III only
 - (d) I and III only
4. Calculate the magnetic field responsible for the polarization of 2.5×10^{-6} for ^{13}C at 298 K.
 5. 0.5 T
 6. 1.2 T
 7. 2.9 T
 8. 100 T

The T_2 of an NMR line is 15 ms. Ignoring other sources of linebroadening, calculate its linewidth.

- (a) 0.0471 Hz
- (b) 21.2 Hz
- (c) 42.4 Hz
- (d) 66.7 Hz

True / False (10 marks)

Answer True or False

6. True or False: All lanthanide elements react with aqueous acid to liberate hydrogen gas in a similar manner to alkali metals.
7. True or False: The NMR frequency of ^{31}P in a 20.0 T magnetic field is 345.0 MHz.
8. True or False: HCl and NaOH are classified as a conjugate acid-base pair in acid-base chemistry.
9. True or False: The ^{13}C spectrum of hexane exhibits three distinct chemical shifts due to its unique molecular structure.
10. True or False: At standard temperature and pressure, bromine (Br_2) exists as a red-brown volatile liquid.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the impact of doubling the volume of a buffer solution, composed of equal concentrations of a weak acid and its conjugate base, on its pH and explain the underlying principles involved.
12. Discuss the factors that contribute to lattice enthalpy in ionic compounds, and analyze why magnesium oxide (MgO) exhibits the greatest lattice enthalpy among common ionic substances.

End of Paper